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Systematic Review

Diagnostic Accuracy of Novel Optical Imaging Techniques for Melanoma Detection: A Systematic Review and Meta-Analysis

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ABSTRACT

The incidence of melanoma is increasing worldwide, requiring early detection to improve survival rates. Although dermoscopy is the standard non-invasive tool for diagnosing melanoma, it relies on experience and skill. Advances in optical imaging technologies and artificial intelligence have the potential to improve diagnostic accuracy. Our objective was to compare the diagnostic accuracy of novel non-invasive optical imaging techniques for melanoma detection. A systematic literature search was conducted in three databases (Medline, Embase, and CENTRAL) on November 15, 2023. Inclusion criteria focused on studies comparing the accuracy of optical imaging methods against histopathology. Outcomes consisted of measures of diagnostic accuracy. Random-effects meta-analyses were performed for each method with 95% confidence intervals to summarize all relevant effect sizes. Of the 16,239 records, 141 articles met the inclusion criteria, of which 138 articles were eligible for the meta-analysis. Reflectance confocal microscopy (RCM) and dermoscopy combined with artificial intelligence (DSC + AI) had the highest sensitivity (0.93), with DSC + AI showing higher specificity (0.77 [0.70–0.83]) than RCM (0.749 [0.7475–0.7504]). Multispectral imaging combined with AI also showed high sensitivity (0.92 [0.82–0.97]) and relatively high specificity (0.80 [0.67–0.89]). Standalone dermoscopy exhibited balanced sensitivity (0.87 [0.84–0.90]) and specificity (0.82 [0.78–0.86]). In melanoma diagnosis, both RCM and DSC + AI can serve as second-step optical evaluation methods for suspicious lesions following initial screening with DSC. By maintaining a strong emphasis on multimodal imaging, healthcare providers could improve early detection and outcomes for patients with melanoma.

1 | Introduction

Melanoma, a malignant tumor arising from melanocytes, is a major public health concern worldwide due to its aggressive nature and high potential for metastasis. The incidence of melanoma has been steadily rising over the past decades [1]. The current gold standard for the diagnosis of melanoma is histopathology [2], although it requires an invasive procedure and does not provide prompt results. Early detection is crucial for reducing the tumor burden [3], as timely initiation of current therapeutic options has shown high efficacy in improving survival rates [4]. However, melanoma represents a significant

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diagnostic challenge due to its heterogeneity in clinical appearance [5].

Dermoscopy (DSC), also known as dermatoscopy or epiluminescence microscopy, is a non-invasive imaging technique that enhances diagnostic accuracy in detecting melanoma and other pigmented skin lesions [6].

Advances in technology have led to the development of novel non-invasive optical imaging techniques for the detection of melanoma [4]. The advantage of these imaging methods compared to conventional DSC is that they may offer higher resolution and often reproducible quantitative measurements [7]. These instruments are based on different biophysical principles, including cameras working with the properties of light and tissues to capture and analyze images at multiple wavelengths (e.g., multispectral imaging (MSI)); analytical techniques that provide information on the molecular composition, structure, and dynamics based on the scattering of photons by molecular vibrations (e.g., Raman spectroscopy (RS)); confocal imaging techniques based on light-tissue interaction (e.g., reflectance confocal microscopy (RCM)); devices operating with light interference to visualize tissue microstructures (e.g., optical coherence tomography (OCT)); and fluorescence methods (e.g., dermatofluoroscopy (DF)) [8]. Teledermoscopy (TeleDSC) is an emerging diagnostic tool for the prevention and diagnosis of skin diseases when face-to-face visualization of lesions is difficult [9]. The evaluation and diagnosis of the obtained images can also be assisted using artificial intelligence (e.g., DSC combined with artificial intelligence (DSC + AI) and multispectral imaging combined with artificial intelligence (MSI + AI)), thereby increasing diagnostic efficacy and objectivity.

This study aims to compare the diagnostic accuracy of novel non-invasive optical imaging techniques for melanoma detection.

2 | Materials and Methods

We conducted a systematic literature review and meta-analysis following the latest guidelines of the Cochrane Handbook [10]. The review protocol was registered in the PROSPERO International Prospective Register of Systematic Reviews (registration number CRD42023480274, see https://www.crd.york.ac.uk/prospero). We did not deviate from the PROSPERO protocol. We reported our systematic review and meta-analysis using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 Statement (Table S1) [11].

2.1 | Search Strategy

We performed a systematic search in three scientific databases: Medline (via PubMed), CENTRAL (The Cochrane Central Register of Controlled Trials), and Embase from inception to November 15, 2023. The search strategy included three key elements: (i) terms related to melanoma, (ii) terms related to diagnosis, and (iii) a list of all optical imaging techniques. We used the predefined search key in all databases without any filters or restrictions. The detailed search strategy can be found in the Supporting Information (Supporting Information Methods).

2.2 | Selection and Eligibility Criteria

Inclusion criteria were defined as any peer-reviewed studies focusing on the diagnostic accuracy of optical imaging devices if they met the following population–index test–reference test–diagnosis (PIRD): (P) patients with suspected melanoma, (I) non-invasive optical diagnostic techniques, (R) histology, and (D) melanoma. There were no restrictions on the age, sex, or ethnicity of the study population included in the articles. We excluded studies in which we could not extract valuable data from the paper or if they included fewer than five melanoma cases, case reports, meta-analyses, systematic reviews, or letters. After removing duplicates using reference management software EndNote 20 (Clarivate Analytics, Philadelphia, PA, USA) [12], two review authors (N.N.V. and L.G.) independently screened titles, abstracts, and then full texts against predefined eligibility criteria to perform the selection. A third review author (N.K.) resolved conflicts.

2.3 | Data Extraction

Two independent review authors (N.N.V. and L.G.) extracted data from eligible studies into a standardized data collection form. An independent third author (N.K.) resolved all disagreements.

2.4 | Risk of Bias Assessment

The risk of bias was assessed based on the recommendation of the Cochrane Collaboration [10]. For diagnostic questions, we used the quality assessment of studies of diagnostic accuracy included in the systematic reviews (QUADAS-2) tool [13]. In each study, we analyzed the risk of bias for every imaging technique separately. Two independent reviewers (N.N.V. and L.G.) performed the quality assessment of the outcomes, and an independent third investigator (N.K.) resolved the disagreements. Publication bias was assessed by visual inspection of funnel plots.

2.5 | GRADE Assessment

We assessed the quality of evidence in the included studies using the GRADE (grading of recommendations assessment, development, and evaluation) [14] approach with the GRADEpro GDT software (GRADEpro GDT: GRADEpro Guideline Development Tool. McMaster University and Evidence Prime, 2024, available from gradepro.org). Outcomes were rated as high, moderate, low, or very low quality of evidence.

2.6 | Statistical Analysis

For the estimation of pooled specificity and sensitivity, the bivariate model of Chu et al. [15] and Reitsma et al. [16] was fitted. This approach considers the possible association between

sensitivity and specificity. We plotted individual and pooled sensitivities and specificities of the studies included, their summary estimates, and the corresponding 95% confidence and prediction regions. In these visualizations, the size of the ellipsoids reflects the weight of the studies calculated according to the method described by Burke et al. [17]. The PPV and NPV results were calculated from the estimated specificity and sensitivity values at a melanoma prevalence rate of 30%, as this became the average prevalence value from the studies examining melanoma suspicious lesions. We also calculated the diagnostic odds ratio (DOR) with its 95% CI, a single indicator that combines the sensitivity and specificity of a diagnostic test, thus simplifying the comparison of test performance. It is defined as the ratio of the odds of a positive test result for individuals with the disease compared to the odds of a positive test result for individuals without the disease. The DOR ranges from 0 to infinity, with higher values indicating better performance [18]. Heterogeneity was assessed by calculating the I^2 measure and its confidence interval arising from the separate univariate analyses. Statistical analyses were performed using R statistical software (version 4.1.2., R Core Team (2023). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria, https://www.r-project.org/) using the meta [19] and the lme4 [20] packages and were partially based on the web tool of Freeman et al. [21]. The statistical analyses followed the advice of Harrer et al. [22].

3 | Results

3.1 | Search and Selection

The PRISMA flowchart [11] is shown in Figure 1. Initially, we identified 16,239 records via database searching, of which 136 studies [23–158] were found eligible. In addition, via citation searching, we found 5 more eligible studies [159–163]. Overall, 141 articles were included in the qualitative analysis (systematic review), of which 138 studies were included in the quantitative synthesis (meta-analysis).

3.2 | Basic Characteristics of Studies Included

The baseline characteristics of the studies included in the meta-analyses are detailed in Table S2.

3.3 | Quantitative Analysis (Meta-Analysis)

The bivariate analyses revealed that different optical imaging techniques exhibited varying levels of sensitivity and specificity (see Figure 2). Among the methods evaluated, DSC + AI and RCM equally showed the highest sensitivity (0.93 [0.88; 0.96] and 0.93 [0.9282; 0.9293], respectively). However, DSC + AI showed a slightly higher specificity (0.77 [0.70; 0.83]) than

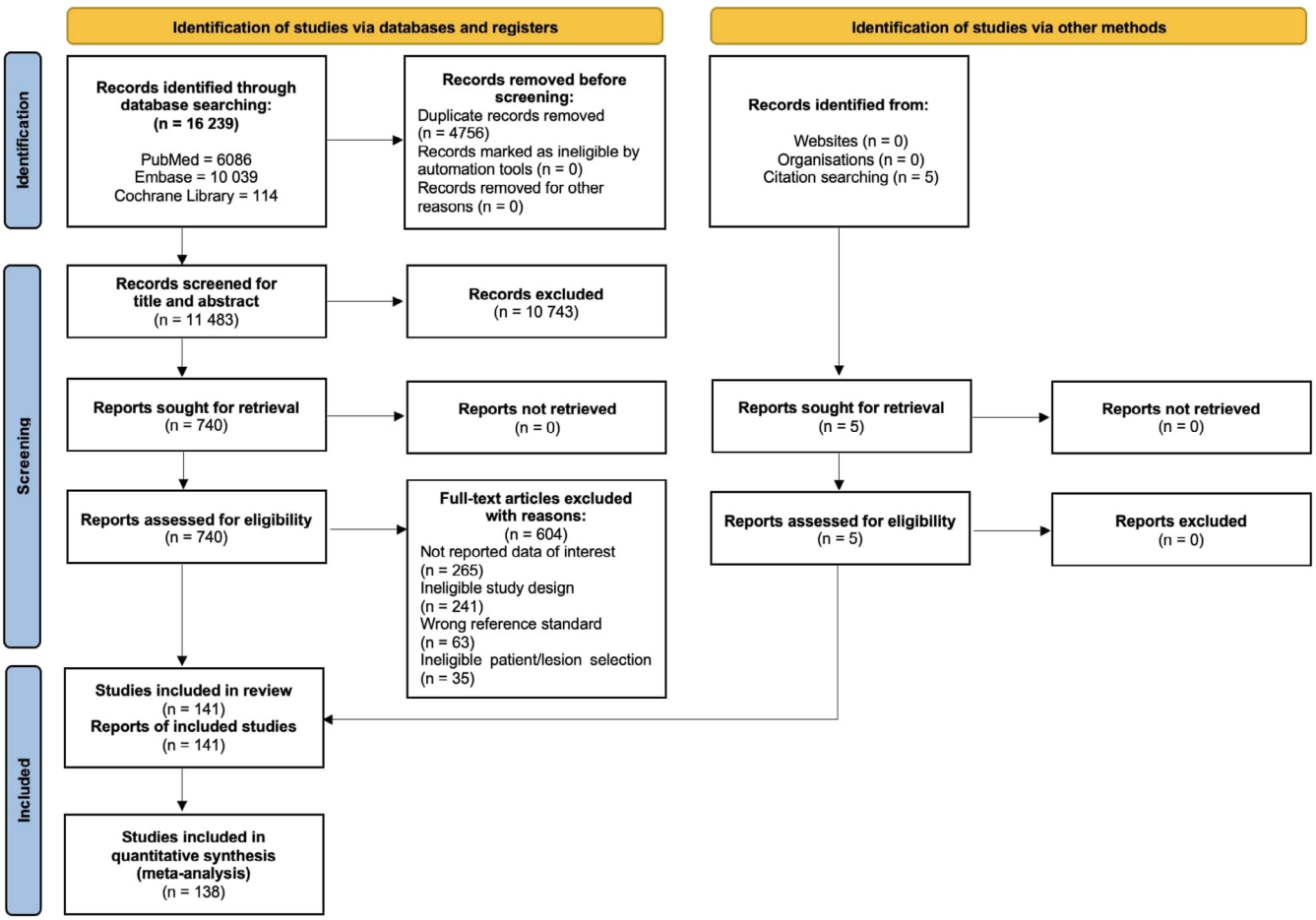


FIGURE 1 | Preferred reporting in systematic reviews and meta-analyses (PRISMA) flowchart showing the selection process.

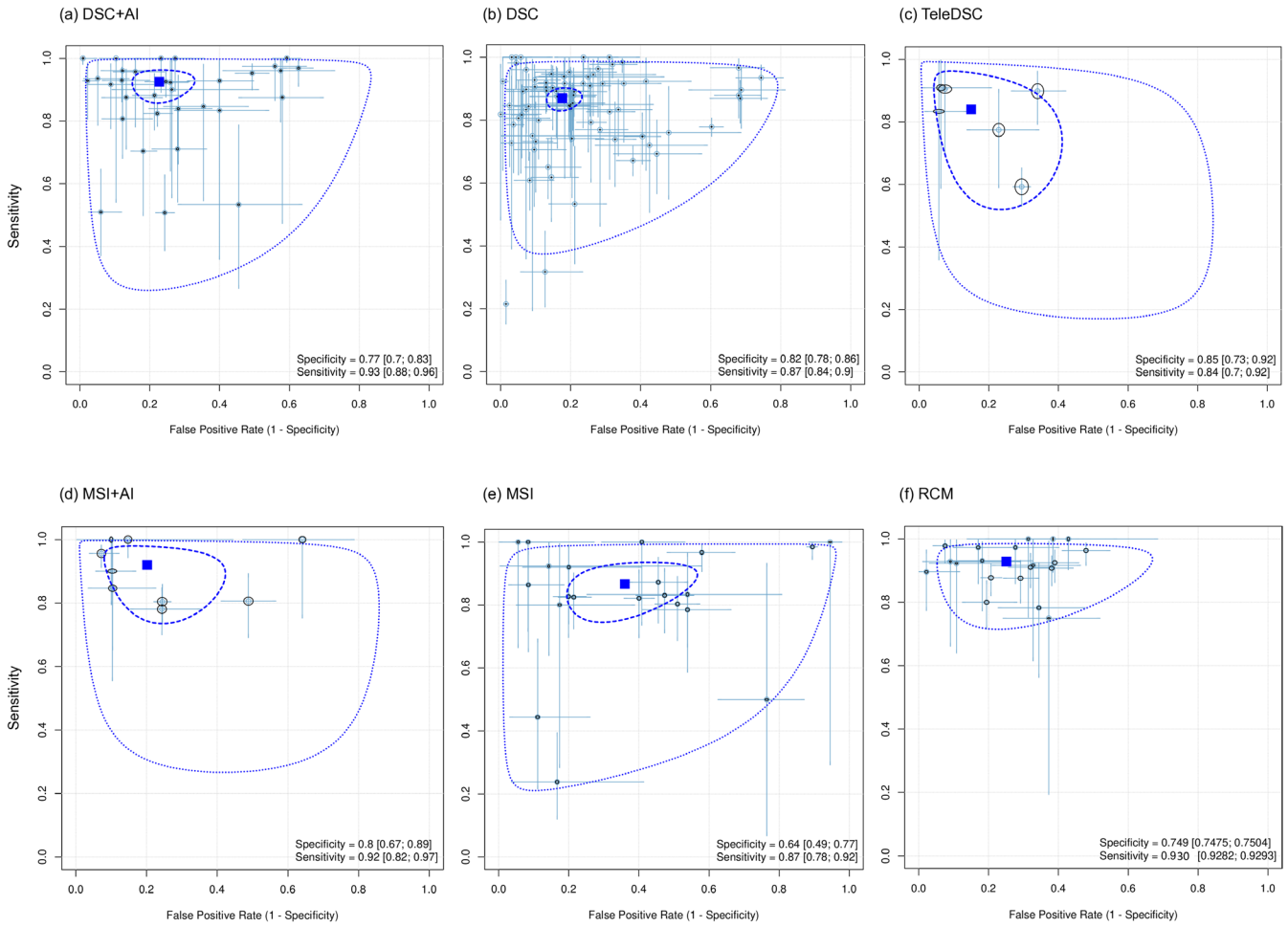


FIGURE 2 | Bivariate plot of the articles included in the meta-analysis evaluating the diagnostic accuracy of (a) DSC + AI, (b) DSC, (c) TeleDSC, (d) MSI + AI, (e) MSI, and (f) RCM. On each figure, the sensitivity is plotted against the false positive rate (1—specificity), with individual study estimates (black circles) and their confidence intervals (blue error bars). The summary points (blue square) represent the pooled sensitivity and specificity with 95% confidence regions (dashed blue ellipse) and sROC curves (dotted blue line). The pooled specificity and sensitivity values are reported in the bottom right corners. DSC, dermoscopy; DSC + AI, dermoscopy + artificial intelligence; MSI, multispectral imaging; MSI + AI, multispectral imaging + artificial intelligence; RCM, reflectance confocal microscopy; sROC, summary receiver operating characteristic; TeleDSC, teledermoscopy.

RCM (0.749 [0.7475; 0.7504]). MSI + AI also demonstrated high sensitivity (0.92 [0.82; 0.97]) and a relatively high specificity (0.80 [0.67; 0.89]). The specificities of standalone DSC and TeleDSC were closer to each other (0.82 [0.78; 0.86] and 0.85 [0.73; 0.92]), though DSC had a higher sensitivity (0.87 [0.84; 0.90]) compared to TeleDSC (0.84 [0.70; 0.92]). MSI, without the use of AI, showed lower specificity (0.64 [0.49; 0.77]) but maintained high sensitivity (0.87 [0.78; 0.92]). The outcomes of these individual summary receiver operating characteristic (sROC) curves, combined into a bivariate random-effects model, are shown in Figure 3. All individual Forest plots for the pooled sensitivity and specificity results of these optical imaging tools can be found in the Supporting Information (Figures S1–S12).

Table 1 summarizes the sensitivity, specificity, PPV, NPV, DOR, and corresponding certainty of the evidence for all optical imaging modalities investigated in the meta-analysis. In addition to the previously mentioned six methods, we investigated three other optical imaging techniques: DF, OCT, and RS. Due to the limited number of articles, we could only perform univariate analyses, the results of which are presented in Table 1. The detailed sensitivity and specificity Forest plots of these three imaging modalities are shown in Figures S13–S18. DSC + AI, RCM, and MSI + AI achieved the highest NPV values (0.96). OCT and TeleDSC reached the highest PPV values (0.71). Overall, OCT demonstrated the highest DOR (64.25 [0.25; 16391.97]) and nearly similar sensitivity (0.84 [0.72; 0.96]) and specificity (0.85 [0.63; 1.00]) results. RCM showed high levels of evidence for sensitivity (0.93 [0.928; 0.93]) and moderate levels of evidence for specificity (0.75 [0.748; 0.75]). Combining AI with imaging modalities such as DSC and MSI generally improved diagnostic performance (DOR of DSC from 27.32 [19.54; 38.20] to 32.62 [17.74; 59.97] and DOR of MSI from 9.62 [4.80; 19.28] to 37.51 [10.16; 138.51]). DSC, TeleDSC, and RS showed moderate to low diagnostic performance (DOR of DSC 27.32 [19.54; 38.20], DOR of TeleDSC 24.00 [4.88; 118.04], DOR of RS 22.66 [0.81; 631.00]). DF, despite its high sensitivity (0.91 [0.88; 0.95]), had the lowest specificity (0.55 [0.37; 0.74]) of the methods evaluated and showed very low evidence levels in both cases. Forest plots of DOR for each optical method can be found in the Supporting Information (Figures S19–S27).

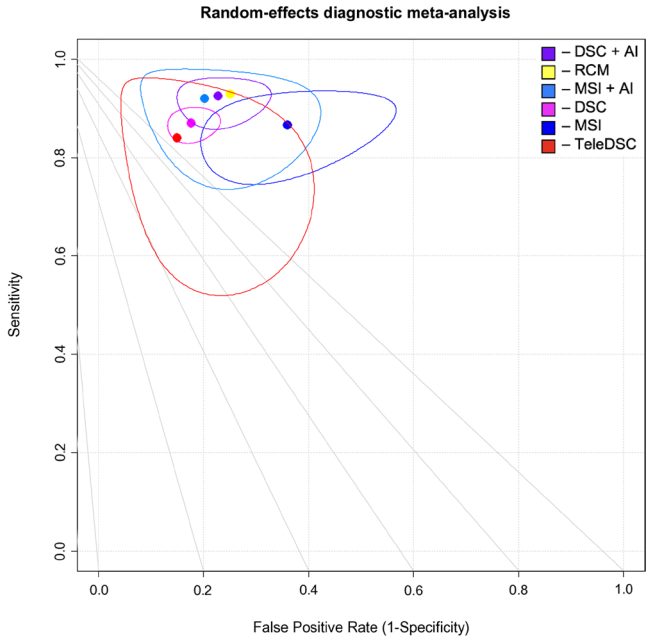


FIGURE 3 | Random-effects diagnostic meta-analysis of optical imaging techniques used in melanoma diagnostics. The sROC curve shows the sensitivity and (1—specificity) of each method, with different colors representing different techniques. Small circles represent summary estimates, and ellipses around circles represent the 95% confidence intervals for every technique. The position of the dots and the size of the ellipses provide a visual comparison of the diagnostic accuracy and reliability of each method. DSC, dermoscopy; DSC + AI, dermoscopy + artificial intelligence; MSI, multispectral imaging; MSI + AI, multispectral imaging + artificial intelligence; RCM, reflectance confocal microscopy; sROC, summary receiver operating characteristic; TeleDSC, teledermoscopy.

3.4 | Qualitative Analysis (Systematic Review)

Of the three studies only included in the systematic review (Table S3), near-infrared and skin impedance (NIR-SI) spectroscopy [161] demonstrated high specificity (0.95) and sensitivity (0.83). Multimodal spectral diagnosis (MSD) [98] achieved a sensitivity and specificity of 1.00. Multiphoton microscopy (MPM) [63] showed moderate sensitivity (0.75 [0.57–0.93]) and specificity (0.80 [0.67–0.93]).

3.5 | Risk of Bias Assessment

Table S4 presents the results of the risk of bias assessment and concerns about the applicability of individual studies. Figure S28 shows the proportion of studies evaluated for risk of bias and applicability concerns across the four QUADAS-2 domains.

3.6 | GRADE Assessment

Table 1 presents the level of evidence for sensitivity and specificity of the optical methods included in the meta-analysis, while Table S3 provides this information for the three imaging methods analyzed only in the systematic review. A detailed GRADE assessment for each optical imaging technique is available in the summary of findings tables (see Tables S5–S15).

3.7 | Publication Bias and Heterogeneity

Funnel plots indicating the possibility of publication bias are found in Figures S29–S46. Visual assessment of the funnel plots implied asymmetry and potential publication bias for both sensitivity and specificity of most of the optical methods, except for OCT and MSI + AI. The meta-analyses showed considerable statistical heterogeneity, except for the sensitivity of RCM ($I^2 = 59\%$, substantial statistical heterogeneity), OCT ($I^2 = 43\%$, moderate statistical heterogeneity), and DF ($I^2 = 0\%$, no important statistical heterogeneity); also the DOR of OCT ($I^2 = 65\%$, substantial statistical heterogeneity), RCM ($I^2 = 59\%$, substantial statistical heterogeneity), MSI ($I^2 = 70\%$, substantial statistical heterogeneity), and DF ($I^2 = 53\%$, substantial statistical heterogeneity). The p values related to I^2 were mostly $p < 0.01$.

4 | Discussion

The NCCN (National Comprehensive Cancer Network) [164] and EADO (European Association of Dermato-Oncology) [165] provide comprehensive guidelines for the diagnosis and management of melanoma, emphasizing the importance of a thorough dermoscopic examination for initial screening. Our meta-analysis demonstrated that DSC provides balanced diagnostic accuracy with near-identical sensitivity and specificity results and an intermediate DOR number, which makes it a reliable and widely available choice for the early detection of melanoma. The moderate level of evidence for the GRADE assessment also supports the reliability of this technique. With these results, DSC remains the leading method for melanoma screening. However, proficiency in DSC and improved diagnostic efficacy require significant specialization and practice. This underscores the need for more objective and standardized second-step diagnostic tools that some novel optical methods can provide.

Our findings from the meta-analysis of a total of 141 studies can strengthen the hypothesis that advances in optical imaging and artificial intelligence technologies have the potential to improve melanoma diagnostics. The accuracy of a diagnostic test is primarily determined by its sensitivity, with RCM and DSC + AI achieving the highest sensitivity values in our study. RCM and DSC + AI also demonstrated the highest NPV values, indicating that these modalities became the most effective in excluding melanoma. RCM showed high DOR results and the highest level of evidence for sensitivity, indicating reliable detection of true positive cases. Based on our results, RCM may be the best choice as a second-step evaluation method after screening with dermoscopy. RCM is not only the most sensitive method for diagnosing melanomas that are not clear-cut on dermoscopy but also the most effective in reducing the number of benign lesions excised [166].

The use of AI in dermatology is an emerging field with promising outcomes. Our meta-analysis suggests that the integration of AI with DSC significantly improves the diagnostic performance

TABLE 1 | Ranking of the optical imaging modalities based on the diagnostic odds ratio.

Method	Sensitivity	Specificity	PPV[a]	NPV[a]	Diagnostic odds ratio	Level of evidence for sensitivity	Level of evidence for specificity
OCT	0.84 [0.72; 0.96]	0.85 [0.63; 1.00]	0.71	0.93	**64.25 [0.25; 16391.97]**	⊕⊕◯◯ Low	⊕◯◯◯ Very low
MSI + AI	0.92 [0.82; 0.97]	0.80 [0.67; 0.89]	0.66	0.96	**37.51 [10.16; 138.51]**	⊕◯◯◯ Very low	⊕◯◯◯ Very low
RCM	0.930 [0.9282; 0.9293]	0.749 [0.7475; 0.7504]	0.61	0.96	**35.27 [20.24; 61.47]**	⊕⊕⊕⊕ High	⊕⊕⊕◯ Moderate
DSC + AI	0.93 [0.88; 0.96]	0.77 [0.70; 0.83]	0.64	0.96	**32.62 [17.74; 59.97]**	⊕⊕⊕◯ Moderate	⊕⊕⊕◯ Moderate
DSC	0.87 [0.84; 0.90]	0.82 [0.78; 0.86]	0.68	0.94	**27.32 [19.54; 38.20]**	⊕⊕⊕◯ Moderate	⊕⊕⊕◯ Moderate
TeleDSC	0.84 [0.70; 0.92]	0.85 [0.73; 0.92]	0.71	0.93	**24.00 [4.88; 118.04]**	⊕⊕◯◯ Low	⊕⊕◯◯ Low
Raman	0.74 [0.40; 1.00]	0.79 [0.66; 0.93]	0.60	0.88	**22.66 [0.81; 631.00]**	⊕⊕◯◯ Low	⊕⊕◯◯ Low
DF	0.91 [0.88; 0.95]	0.55 [0.37; 0.74]	0.47	0.94	**11.17 [1.64; 76.24]**	⊕◯◯◯ Very low	⊕◯◯◯ Very low
MSI	0.87 [0.78; 0.92]	0.64 [0.49; 0.77]	0.51	0.92	**9.62 [4.80; 19.28]**	⊕⊕◯◯ Low	⊕◯◯◯ Very low

Abbreviations: DF, dermatofluoroscopy; DSC, dermoscopy; DSC + AI, dermoscopy + artificial intelligence; MSI, multispectral imaging; MSI + AI, multispectral imaging + artificial intelligence; NPV, negative predictive value; OCT, optical coherence tomography; PPV, positive predictive value; RCM, reflectance confocal microscopy; TeleDSC, teledermoscopy.
[a]Average PPV and NPV results at a 30% prevalence rate of melanoma.

of melanoma detection. This correlates with the findings of Salinas et al. [167] that the diagnostic performance of AI algorithms was better than that of non-expert dermatologists and comparable to that of expert dermatologists using DSC alone in clinical practice. In our opinion, DSC + AI is currently best suited for confirming melanoma diagnosis, but in the future, it may surpass standalone DSC in screening as well.

We found similar results when combining AI with other optical imaging modalities, such as MSI. MSI + AI achieved much higher sensitivity, specificity, and DOR values than MSI alone. MSI + AI achieved the highest NPV, suggesting that this method is also reliable in excluding melanoma. DF, despite its high sensitivity, had the lowest specificity and showed very low DOR and evidence levels, suggesting limited diagnostic reliability. Although OCT showed the highest DOR of all the modalities assessed, its sensitivity was only moderate, and the level of evidence supporting the results was also low. TeleDSC, although useful for remote diagnostics, showed lower DOR and overall performance compared to most in-person techniques in our meta-analysis. Although PPVs were generally lower than NPVs, OCT and TeleDSC achieved the highest PPV results, indicating that these modalities are the most likely to confirm melanoma diagnosis. RS achieved comparatively low performance. Its sensitivity, specificity, and DOR results were below those of the other methods, with very low evidence levels, indicating poorer diagnostic power. Although MSD was shown to achieve 100% sensitivity and specificity, the very low level of evidence highlights that further validation is needed to interpret these results. NIR-SI spectroscopy and MPM demonstrated decent sensitivity and specificity results, suggesting their potential in specific clinical scenarios, but further validation and research are needed to confirm their clinical utility. Although these new optical modalities can provide non-invasive and higher-resolution imaging that improves the diagnosis of melanoma, these devices are often expensive and are only available in specialized centers, restricting access to screening. In addition, the interpretation of images requires a high level of expertise, further limiting their use in routine clinical practice [168].

5 | Strengths and Limitations

To our knowledge, this is the first systematic review and meta-analysis to provide a comprehensive analysis of all non-invasive optical imaging techniques for the diagnosis of melanoma. Moreover, the strength of our study lies in its large sample size, encompassing a significant number of patients and lesions. We strictly followed our pre-registered PROSPERO protocol. A rigorous methodology was applied to achieve a high quality of evidence. Both univariate and multivariate analyses were performed to achieve objectivity. In terms of limitations of this work, the significant methodological heterogeneity of the articles included restricted the systematic comparison of all diagnostic techniques, which may affect the consistency and reliability of the results. Furthermore, differences in study types, testing strategy, and imaging device settings may also increase heterogeneity. The presence of a high risk of bias in some of the domains and low levels of evidence for certain optical techniques are also important limitations.

6 | Implications for Practice

Implementing scientific findings into practical recommendations is crucial [169, 170]. First, medical practitioners and dermatologists should prioritize training and skill development in DSC to enhance diagnostic accuracy in melanoma screening. Second, in major centers, clinicians should also consider incorporating multimodal imaging techniques into routine clinical practice after screening with DSC. This approach may provide a more comprehensive evaluation of suspicious pigmented lesions, possibly leading to more accurate diagnoses. This could mean that the introduction of more precise and reliable techniques as second-step evaluation methods can lead to earlier detection of melanoma, accelerate therapy, and improve the survival rates of patients.

7 | Implications for Research

A promising future direction for research can be the investigation of the integration of AI and optical imaging techniques. Studies should focus on developing robust AI models trained on various optical images and evaluating their diagnostic performance. Further research should also explore the efficacy of multimodal imaging in clinical settings for melanoma diagnostics and whether it could synergistically improve diagnostic accuracy. Investigating the impact of multimodal imaging on clinical workflow, feasibility, patient preferences, and cost-effectiveness is also crucial.

8 | Conclusions

Our systematic review and meta-analysis revealed that the reliability and wide availability of DSC still make it the leading optical method for melanoma screening. RCM and DSC + AI may both be utilized as second-step optical evaluation methods of pigmented lesions initially suspicious under DSC. The development of multimodal imaging techniques that complement DSC could potentially maximize the diagnostic accuracy of melanoma detection. By maintaining a strong emphasis on multimodal imaging, healthcare providers could improve early detection and outcomes for patients with melanoma.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used in this study can be found in the full-text articles included in the systematic review and meta-analysis. Data from the article will be shared upon reasonable request to the corresponding author.

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Supporting Information

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